

Team Name

Captain Nemo

Team Member Details

**Atiqur Rahman(Team Leader),
Dania University College, Dhaka, Bangladesh,
Class – 11,
Email: *aatiqurrahman111@gmail.com***

Project: Smart Health Care System

Covid is a contagious disease. As a result, doctors treating corona patients are at greater risk. Moreover, it is often seen that after taking the patient to a hospital, it is known that there is no empty seat to keep the patient in that hospital. As a result, many patients die without getting treatment at the right time. To solve this problem, I have created an eco-system named "**Smart HealthCare System**" between doctors, hospitals and patients.

Project Goals:

- To keep the doctors Safe from Covid or other Contagious disease.
- Giving an accurate idea about the number of hospital seats to patient's relatives.
- Giving a real-time update of a patient to Doctors and the Patient's relatives.
- Make the HealthCare System Smarter.

Video Link: <https://drive.google.com/file/d/14X8MrkCedmR14uWqYSz8lDn1wkYSNGVP/view?usp=sharing>

Admin App

Register a New Hospital

Hospital Registration

XYZ HealthCare

xyz

123456

Register

Already Registered? Login

Register a New Hospital by clicking the register button

If already registered then click the Already Registered button

Hospital Registration

Hospital Name:

Hospital UserName:

Hospital ID:

Register

Already Registered? Login

Click this button if the hospital is already registered.This will take you to the Login Page

Login to the Hospital

Hospital Login

xyz

123456

Login

Don't have an Account? Register

Click the Login button for log into the hospital

Hospital Admin Dashboard

Hospital Admin DashBoard

Add Bed

Edit Bed

Bed Details

Logout

Add a New Bed

Add Bed

Room Number:

Room ID:

Add

All Beds

All Beds

Bed Number - 101

Bed Number - 102

Bed Number - 103

Bed Number - 104

Bed Number - 105

Bed Number - 108

Bed Number - 110

Edit Bed

Edit Bed

Patient Name:

Patient Age:

Patient Gender:

Patient Guardians Phone:

Doctor Name:

Doctor Phone:

Publish

Release Patient

Beds Details

Beds Details

Total Beds

300

Avaiable Beds

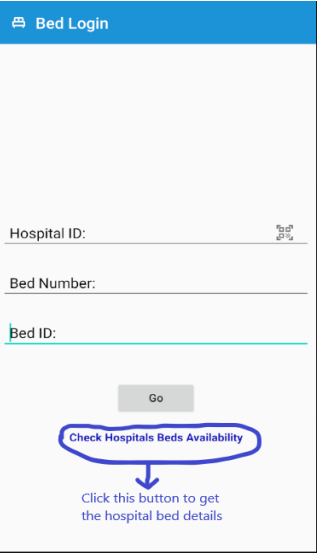
40

Update

This app will Control by the Hospital Authority. In this app Authority can add all the necessary data.

User App

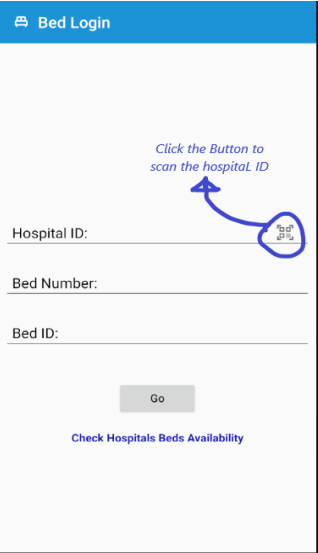
To get the accurate idea about hospital beds click the marked button



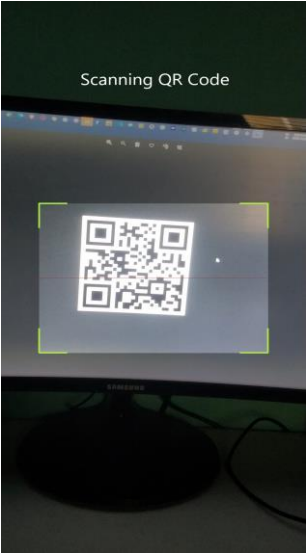
Bed Details



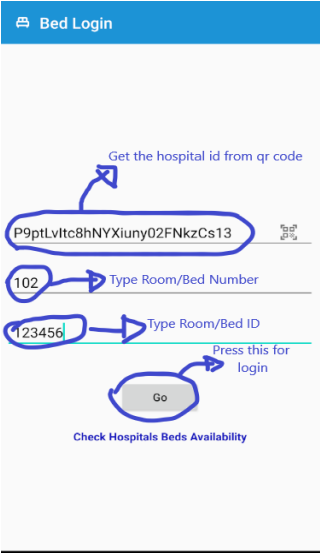
Click the QR Scan Button to get the hospital ID



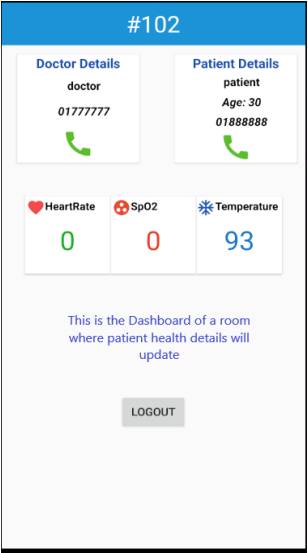
Scanning QR Code



Login to the Bed

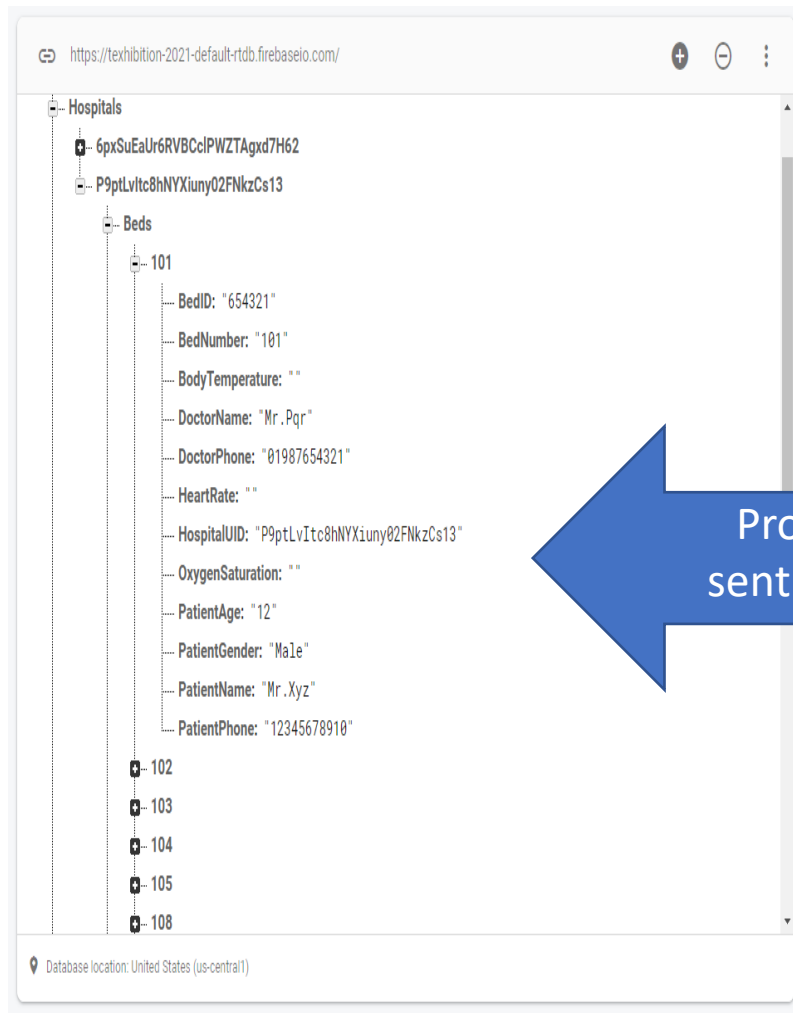


Bed Dashboard

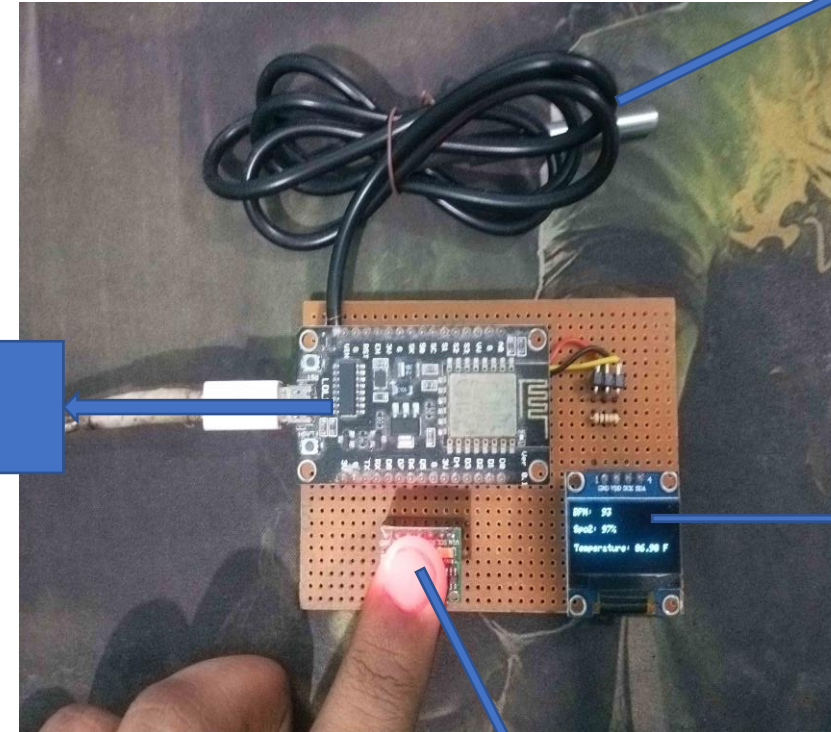


This app will Control by the Doctor and Patient's relatives. In this app they will get the real-time update of that Patient.

Firestore



Process the data & sent to Firebase Cloud



Reading Temperature

Showing Temperature Heart Rate and SPo2

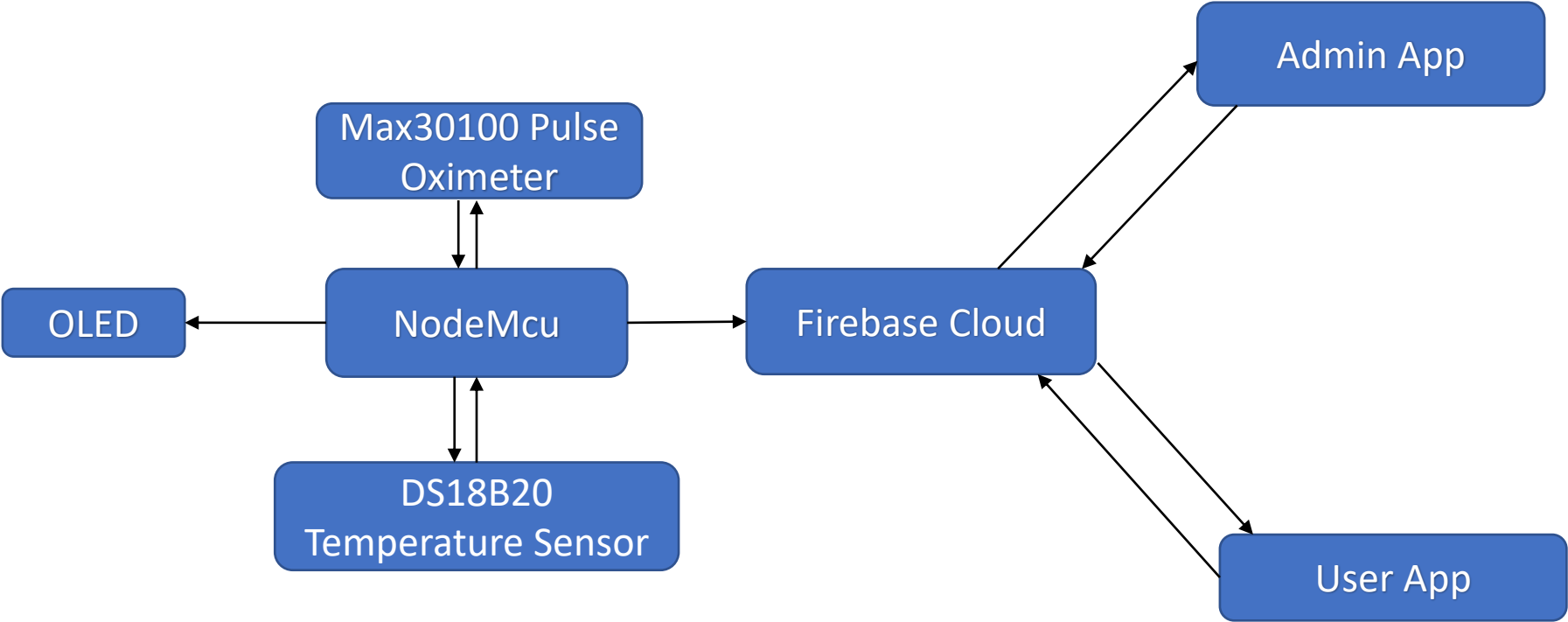
Reading Heart Rate and SPo2

Component Details:

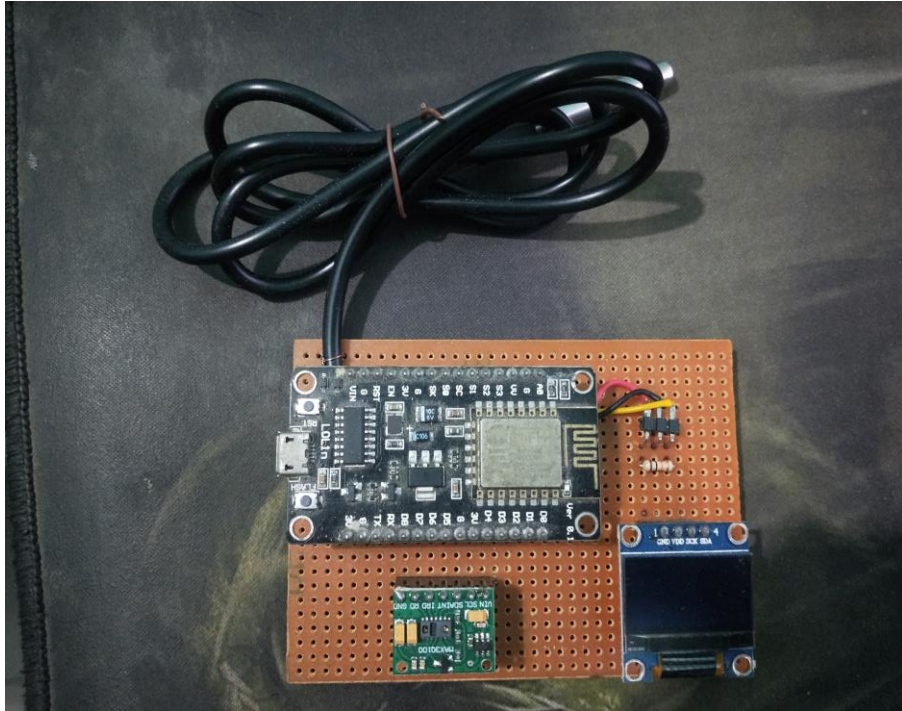
- **NodeMcu ESP8266:** This is the Brain of whole system. It has built-in WIFI. Here we used it's I2C bus for MAX30100 Pulse Oximeter & OLED Display and use one Digital pin for DS18B20 Temperature Sensor. By it's built-in WIFI feature I can send the Heartrate, Oxygen Saturation & Temperature data to the Firebase Cloud.
- **OLED Display Module:** Small in size (0.96") and yet capable OLED display module. It has 128*64 pixel which can display up to 160 characters in total. . It can be used to show real time sensor data from NodeMcu.
- **MAX30100 Pulse-Oximeter:** The MAX30100 is an integrated pulse oximetry and heart-rate monitor sensor solution. It combines two LEDs, a photodetector, optimized optics, and low-noise analog signal processing to detect pulse oximetry and heart-rate signals. This give me the perfect Heart Rate in BPM and Oxygen Saturation level in percentage.
- **DS18B20 Temperature Sensor:** The DS18B20 is one type of temperature sensor and it supplies 9-bit to 12-bit readings of temperature. These values show the temperature of a particular device. The communication of this sensor can be done through a one-wire bus protocol which uses one data line to communicate with an inner microprocessor.



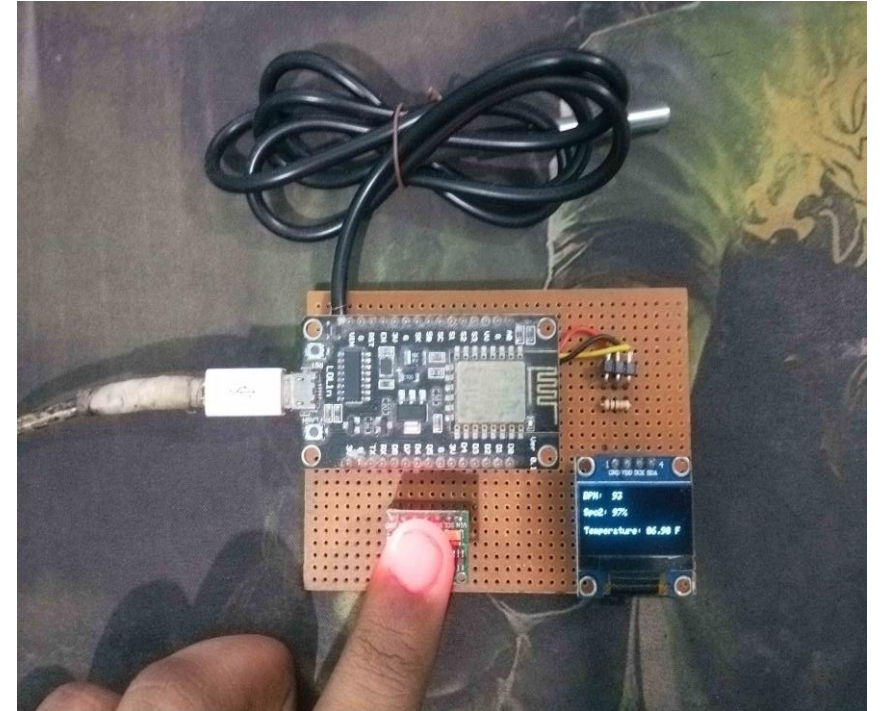
Project Working Diagram



Project Images



Power OFF



Power ON

Key Features

- Patient's relatives will get the accurate information about Hospital Seats.
- Both apps are user friendly.
- Doctor and Patient's relatives can monitor the patient's physical health through the app.
- Patient's heart rate, Spo2 & Body Temperature data will update every 10 second.
- Doctors will be Safe from Covid or other contagious disease.
- Only in case of emergency, the doctor will visit the patient's cabin/room and observe.
- This will make the HealthCare System Smarter.
- Instant Phone Call System.